In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import json
```

In [10]:
```
df = pd.read_csv(r"C:\Users\Sanskruti\OneDrive\Desktop\2019-Nov.csv", nrows=5000
df.head()
```

Out[10]:

	event_time	event_type	product_id	category_id	category_code
0	2019-11-01 00:00:00 UTC	view	1003461	2053013555631882655	electronics.smartphone
1	2019-11-01 00:00:00 UTC	view	5000088	2053013566100866035	appliances.sewing_machine
2	2019-11-01 00:00:01 UTC	view	17302664	2053013553853497655	NaN
3	2019-11-01 00:00:01 UTC	view	3601530	2053013563810775923	appliances.kitchen.washer
4	2019-11-01 00:00:01 UTC	view	1004775	2053013555631882655	electronics.smartphone

In [11]:
```
df.shape
```

Out[11]:
```
(500000, 9)
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['event_time', 'event_type', 'product_id', 'category_id',
       'category_code', 'brand', 'price', 'user_id', 'user_session'],
      dtype='object')
```

In [13]:
```
df['event_type'].value_counts()
```

Out[13]:
```
event_type
view        482642
purchase      9595
cart          7763
Name: count, dtype: int64
```

In [14]:
```
df = df.dropna(subset=['user_id'])
df = df[df['event_type'].isin(['view', 'cart', 'purchase'])]
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 9 columns):
 #   Column         Non-Null Count   Dtype  
---  ------         --------------   -----  
 0   event_time     500000 non-null  object 
 1   event_type     500000 non-null  object 
 2   product_id     500000 non-null  int64  
 3   category_id    500000 non-null  int64  
 4   category_code  337628 non-null  object 
 5   brand          425209 non-null  object 
 6   price          500000 non-null  float64
 7   user_id        500000 non-null  int64  
 8   user_session   500000 non-null  object 
dtypes: float64(1), int64(3), object(5)
memory usage: 34.3+ MB
```

In [15]:
```
visitors = df[df['event_type'] == 'view']['user_id'].nunique()
leads = df[df['event_type'] == 'cart']['user_id'].nunique()
customers = df[df['event_type'] == 'purchase']['user_id'].nunique()

visitors, leads, customers
```

Out[15]:
```
(92743, 4501, 7389)
```

In [16]:
```
visitor_to_lead = (leads / visitors) * 100
lead_to_customer = (customers / leads) * 100
overall_conversion = (customers / visitors) * 100

visitor_to_lead, lead_to_customer, overall_conversion
```

Out[16]:
```
(4.853196467657937, 164.16351921795155, 7.967178115868583)
```

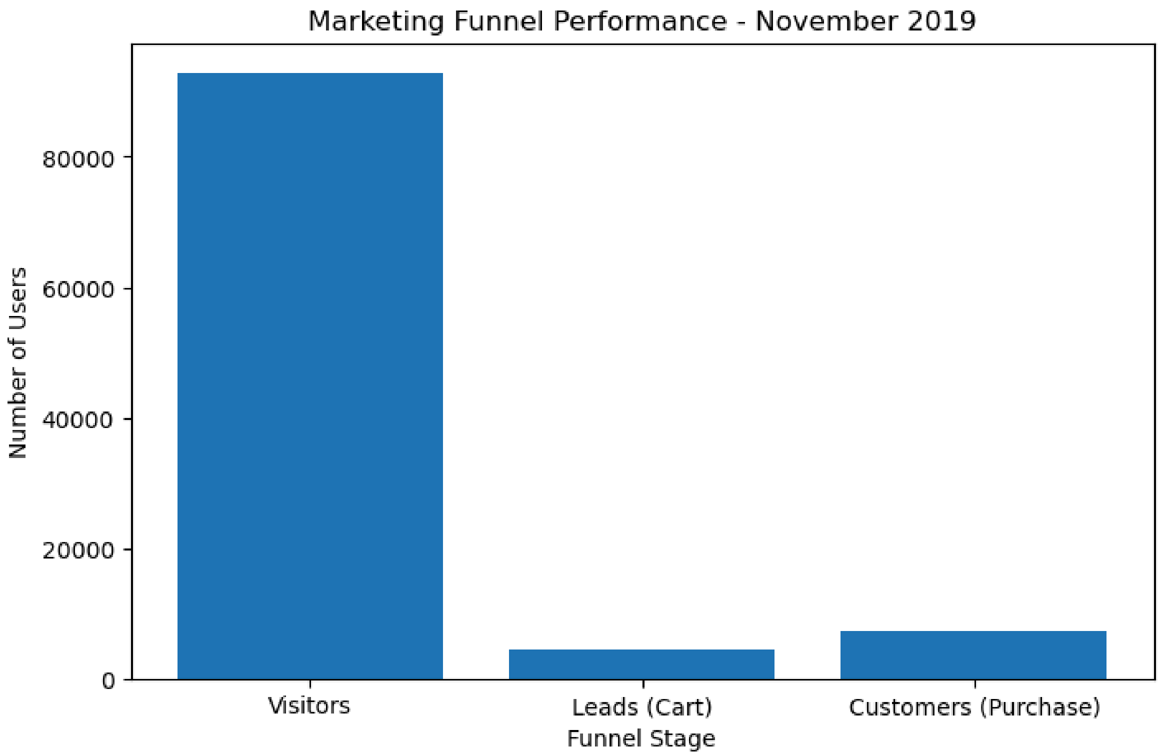
In [17]:
```
funnel_df = pd.DataFrame({
    'Funnel Stage': ['Visitors', 'Leads (Cart)', 'Customers (Purchase)'],
    'Users': [visitors, leads, customers]
})

funnel_df
```

Out[17]:

	Funnel Stage	Users
0	Visitors	92743
1	Leads (Cart)	4501
2	Customers (Purchase)	7389

In [18]:
```
plt.figure(figsize=(8,5))
plt.bar(funnel_df['Funnel Stage'], funnel_df['Users'])
plt.title('Marketing Funnel Performance - November 2019')
plt.xlabel('Funnel Stage')
plt.ylabel('Number of Users')
plt.show()
```



In [19]:

```
drop_visitor_lead = visitors - leads
drop_lead_customer = leads - customers

drop_df = pd.DataFrame({
    'Stage': ['Visitor → Lead', 'Lead → Customer'],
    'Users Dropped': [drop_visitor_lead, drop_lead_customer]
})

drop_df
```

Out[19]:

	Stage	Users Dropped
0	Visitor → Lead	88242
1	Lead → Customer	-2888

In [20]:

```
purchase_df = df[df['event_type'] == 'purchase']

total_revenue = purchase_df['price'].sum()
avg_order_value = purchase_df['price'].mean()

total_revenue, avg_order_value
```

Out[20]:

```
(np.float64(2897621.42), np.float64(301.99285252735797))
```

In [21]:

```
print(f"Visitor to Lead Conversion Rate: {visitor_to_lead:.2f}%")
print(f"Lead to Customer Conversion Rate: {lead_to_customer:.2f}%")
print(f"Overall Conversion Rate: {overall_conversion:.2f}%")
print(f"Total Revenue: {total_revenue:.2f}")
print(f"Average Order Value: {avg_order_value:.2f}")
```

```
Visitor to Lead Conversion Rate: 4.85%
Lead to Customer Conversion Rate: 164.16%
Overall Conversion Rate: 7.97%
Total Revenue: 2897621.42
Average Order Value: 301.99
```

In [ ]: